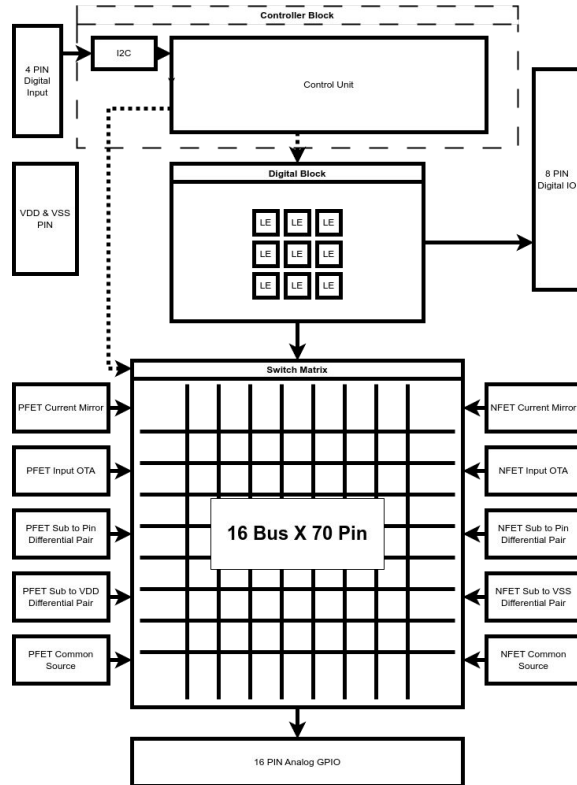


M6_Low Noise Anomaly



Mixed-Signal MOSbius Style:

4x Programmable Digital CLB

1x P-OTA

1x N-OTA

4x PFET Current Mirror

4x NFET Current Mirror

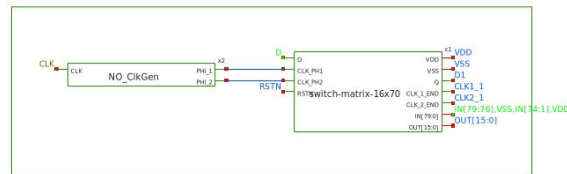
8x Individual PFET

8x Individual NFET

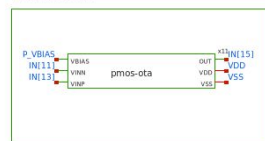
M6_Low Noise Anomaly

Top-Level Module

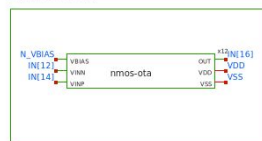
SWITCH MATRIX 16 BUS & 70 PIN



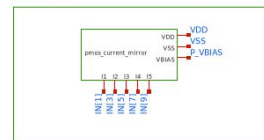
PMOS-5T-OTA



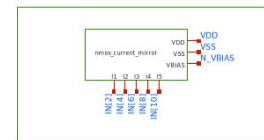
NMOS-5T-OTA



PMOS-CM



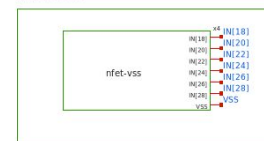
NMOS-CM



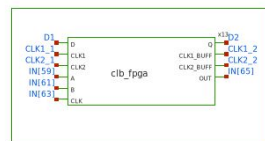
PMOS-VDD



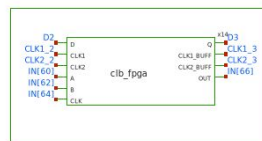
NMOS-VSS



CLB 1



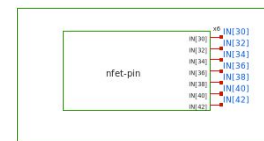
CLB 2



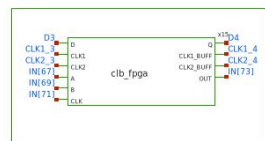
PMOS-PIN



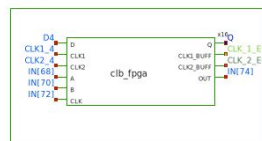
NMOS-PIN



CLB 3



CLB 4



PMOS-COMMON



NMOS-COMMON



std-cell =
gf180mcu_fd_sc_mcu9t5v0

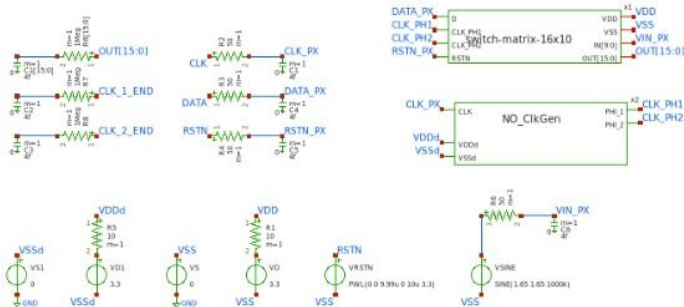
VDD = 3v3

Switch-matrix size =
75 Pin X 16 Bus



Mini TB top-level Simulations

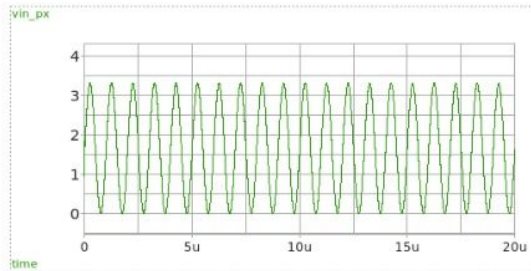
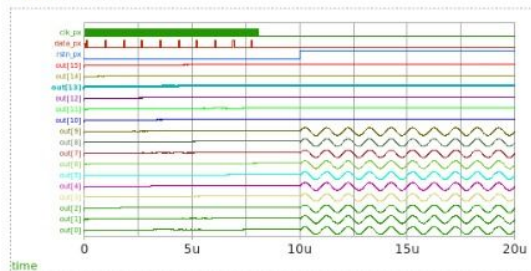
M6_Low Noise Anomaly : Connection test W/ MOSbius Architecture



The Switch Bit-Stream

```
=====
1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0
0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0
0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0
0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0
0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0
0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0
0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0
0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0
```

load waves



```

.NETLIST
global VDDd VSSd

* clock
abit [ bit_node ] input_vector
.model input_vector d_source[input_file="/foss/designs/sscs-chipathen2025-LNA/designs/tb_top-level/tb_mini-top-level/data.txt"]
* data
aclock [ clock_node ] clock_vector
.model clock_vector d_source[input_file="/foss/designs/sscs-chipathen2025-LNA/designs/tb_top-level/tb_mini-top-level/data_clk.txt"]
* convert digital signals to analog
aconvert [ bit_node clock_node ] data clk in
.model dac in dac_bridge[out_low=0V out_high=3.3V t_rise=0.2ns t_fall=0.2ns]

Simulation
.option wnflag=0 bypass=1
.options method=trap rawfile=binary
.options solver=klu nomod

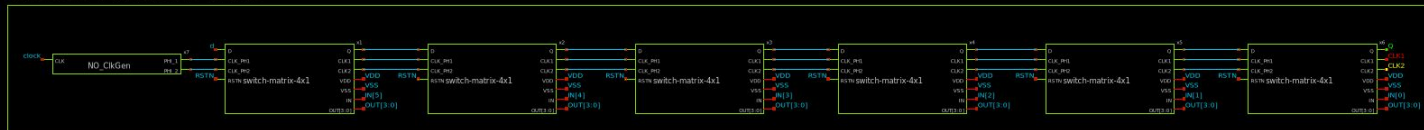
.control
set num threads=6
save VIN_PX 'OUT[0]' 'OUT[1]' 'OUT[2]' 'OUT[3]' 'OUT[4]' 'OUT[5]' 'OUT[6]' 'OUT[7]'
+ 'OUT[8]' 'OUT[9]' 'OUT[10]' 'OUT[11]' 'OUT[12]' 'OUT[13]' 'OUT[14]' 'OUT[15]'
+ CLK_PX DATA_PX RSTN_PX
TRAN 10n 20u uic
write tb_mini-top-level.raw

Models
.endc

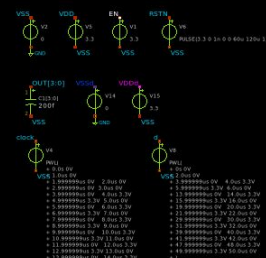
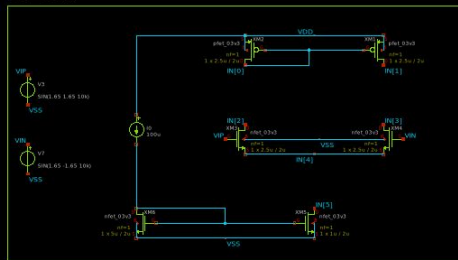
```

M6_Low Noise Anomaly : Single FET test W/ MOSbius Architecture

SWITCH MATRIX AND 4 BUS



5T-OTA



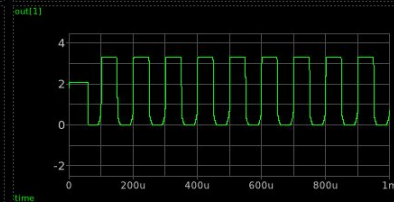
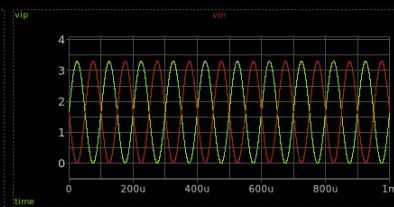
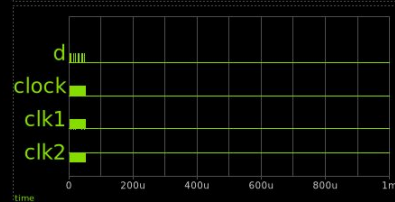
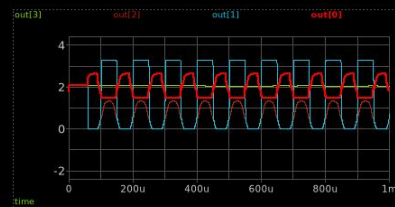
➡ load waves

MODELS

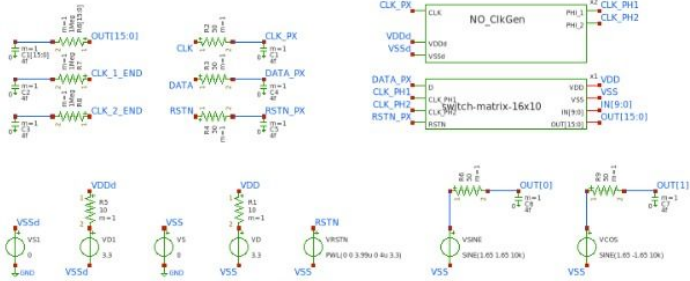
```
.include $:/lib/ncu/Models/design.ngspice
lib $:/lib/ncu/Models/typical
include /foss/pdks/gf180mcu0/lib/ncu.ref/gf180mcu0/spice/gf180mcu0/sc_mcu0t5v0.spice

[
option wflap0 bypass=1
options methodtrap rawfilebinary
options solver=lsf memod
global VDD0 VSS0

.tran 100n 1m
.control
save all
run
plot V(clock)
write tb_switch_matrix_tg_array.dxl.raw
quit 0
endc
```



Architecture



```

.NETLIST

global VDDd VSSd

* clock
abit [bit_node] input_vector
.model input_vector d_source input_file="/foss/designs/sscs-chipathon2025-LNA/designs/tb_top-level/tb_mini-ota-top-level\data.txt")
* data
clock [clock_node] clock_vector
.model clock_vector d_source input_file="/foss/designs/sscs-chipathon2025-LNA/designs/tb_top-level/tb_mini-ota-top-level\data_clk.txt")
* convert digital signals to analog
convert [bit_node clock_node] [data clk] dac_in
.model dac_in dac_bridge [out_low=0V out_high=3.3V t_rise=0.2ns t_fall=0.2ns]

Simulation

.option wflag=0 bypass=1
.options method=trap rawfile=binary
.options solver=klu nomod

.control
set num_threads=6
save 'OUT[0]' 'OUT[1]' 'OUT[2]' 'OUT[3]' 'OUT[4]' 'OUT[5]' 'OUT[6]' 'OUT[7]'
+ 'OUT[8]' 'OUT[9]' 'OUT[10]' 'OUT[11]' 'OUT[12]' 'OUT[13]' 'OUT[14]' 'OUT[15]'
+ CLK_PX DATA_PX RSTN_PX
TRAN 2n 200u uic
write tb_mini-ota-top-level.raw

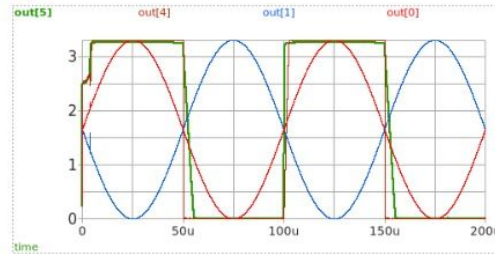
Model

.include /foss/pdks/gf180mcu0/Libs.ref/gf180mcu_fd_sc_mcu9t5v0/spice/gf180mcu_fd_sc_mcu9t5v0.spice
.include $:/180MCU_MODELS/design.ngspice
.lib $:/180MCU_MODELS/sml41064.ngspice typical
.lib $:/180MCU_MODELS/sml41064.ngspice cap_min
.lib $:/180MCU_MODELS/sml41064.ngspice res_typical
.lib $:/180MCU_MODELS/sml41064.ngspice nmoscap_typical
.lib $:/180MCU_MODELS/sml41064.ngspice mimoscap_typical

```

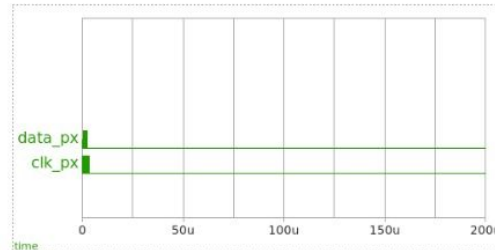


→ load waves

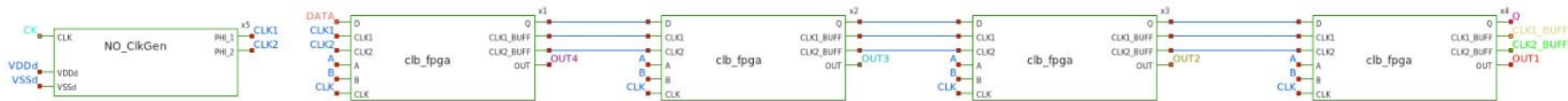


The Switch Bit-Stream

```
=====
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0
0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
```



M6_Low Noise Anomaly : CLB (Digital) test W/ MOSbius Architecture



➔ load waves

Models

```
.include /foss/pdks/gf180mcuD/libs.ref/gf180mcu_fd_sc_mcu9t5v0/spice/gf180mcu_fd_sc_mcu9t5v0.spice
.include $:180MCU_MODELS/design.ngspice
.lib $:180MCU_MODELS/sm141064.ngspice typical
```

Simulation

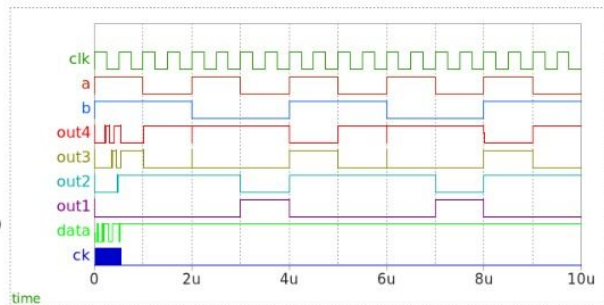
```
.global VDDd VSSd
* clock
ablt [ bit_node ] input_vector
.model input_vector d_source(input_file="/foss/designs/sscs-chipathon2025-LNA/designs/libs/tb_fpga/tb_clb/data.txt")
* data
aclock [ clock_node ] clock vector
.model clock_vector d_source(input_file="/foss/designs/sscs-chipathon2025-LNA/designs/libs/tb_fpga/tb_clb/data_clk.txt")
* convert digital signals to analog
aconvert [ bit_node clock_node ] [ data ck ] dac in
.model dac_in dac_bridge (out_low=0V out_high=3.3V t_rise=0.2ns t_fall=0.2ns)

.control
save all
TRAN 0.2n 10u
write tb_clb.raw
.endc
```

CLB Register

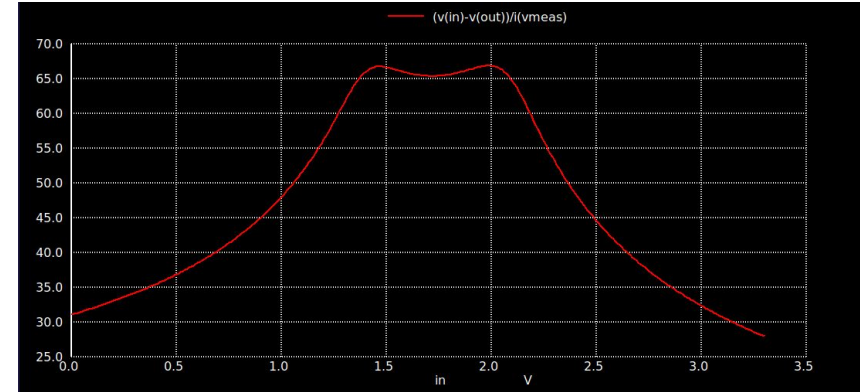
```
1000,[0/1] => NAND
0111,[0/1] => OR
0001,[0/1] => NOR
1110,[0/1] => AND
1001,[0/1] => XNOR
0110,[0/1] => XOR
```

NB : [0/1] is MSB stand for [0/1] => [Sequential(FF)/Combinational(ABC)]



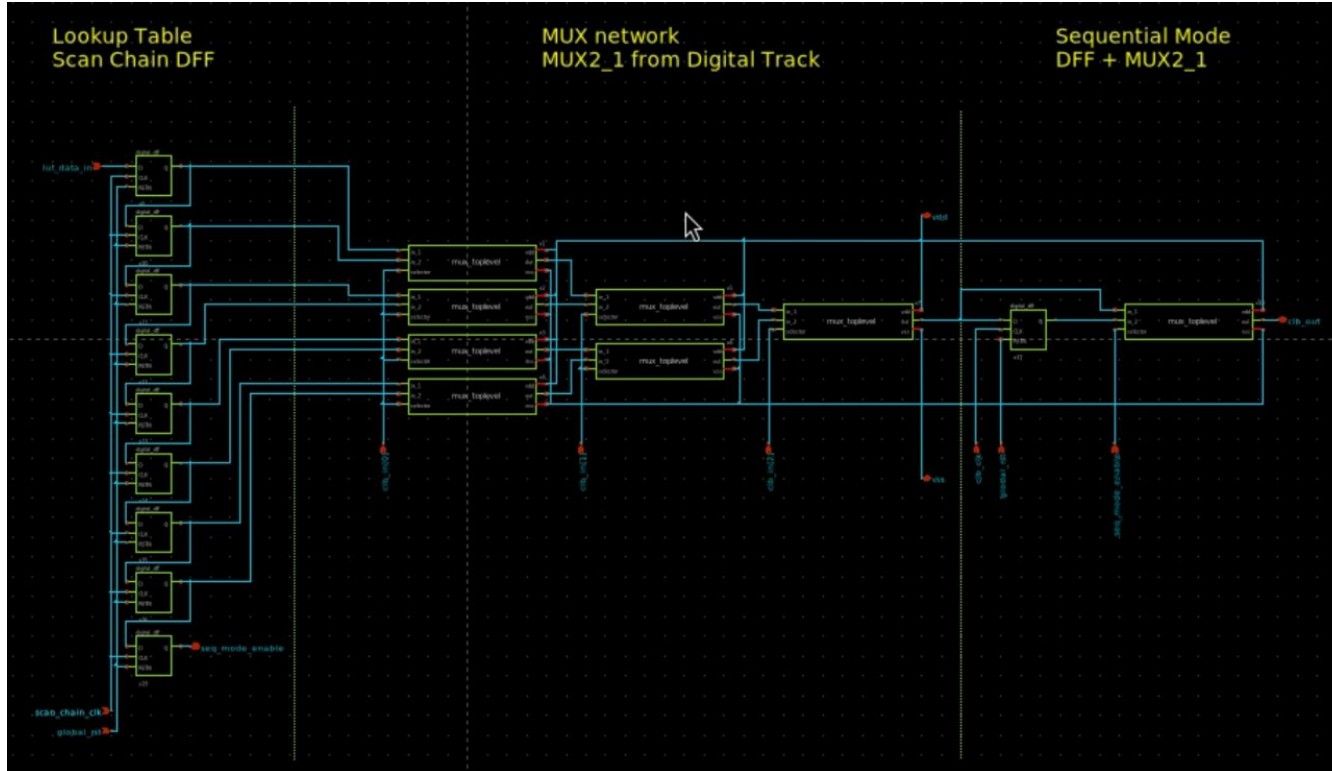
Spec and Test Every
Module

Switch-matrix Transmission Gate Specs

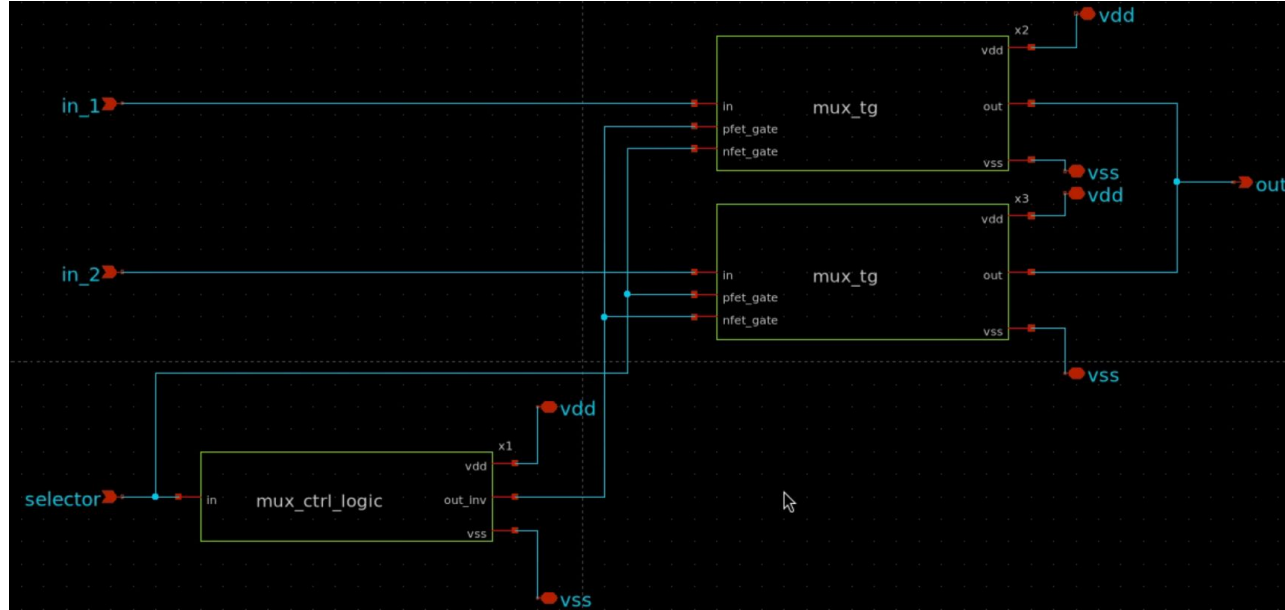


Already achieve < 100 Ohm each Switch Cell,
[Spec is MET]

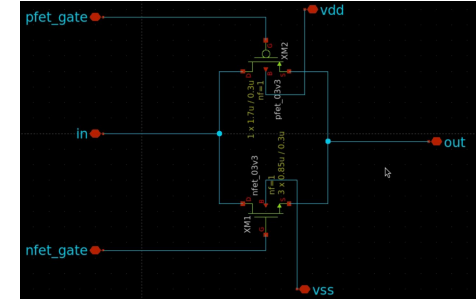
M6_Low Noise Anomaly : CLB Schematic



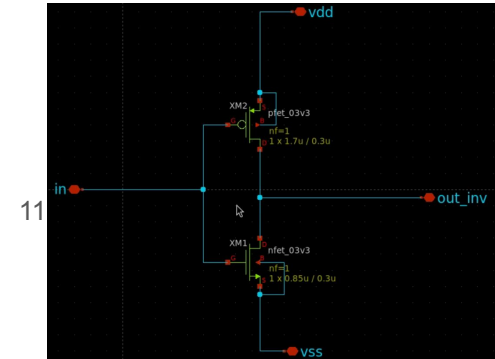
M6_Low Noise Anomaly : CLB Internals - MUX2_1



Top Level Schematic for MUX2_1

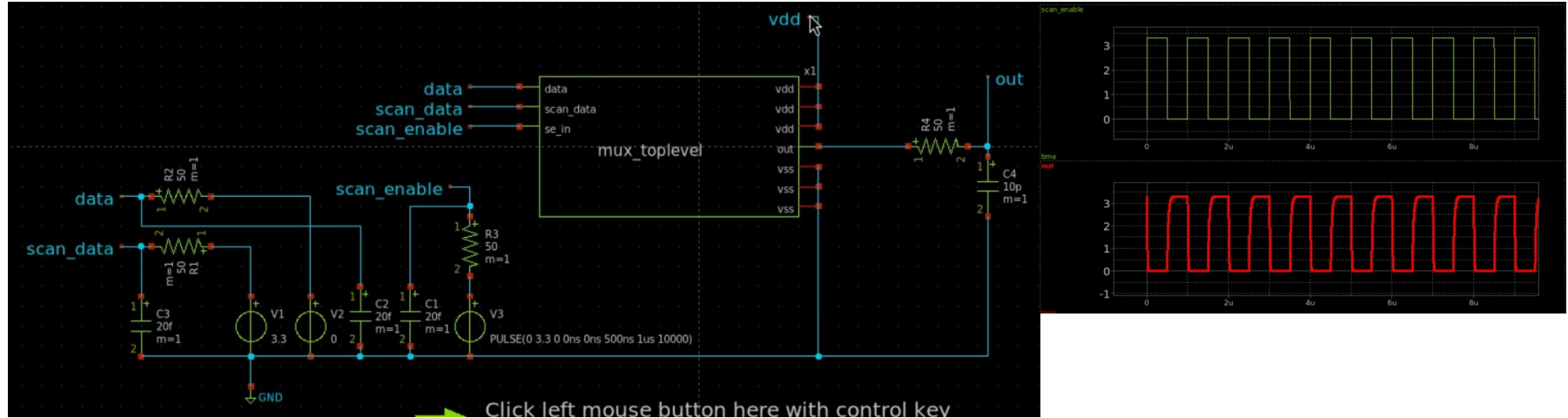


Transmission Gate



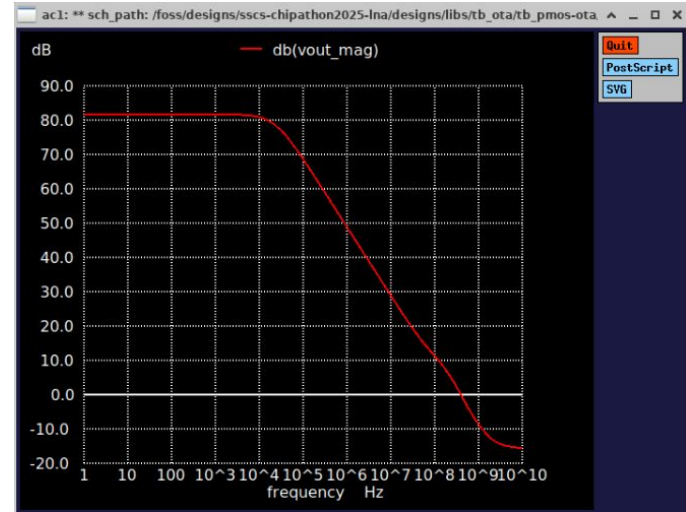
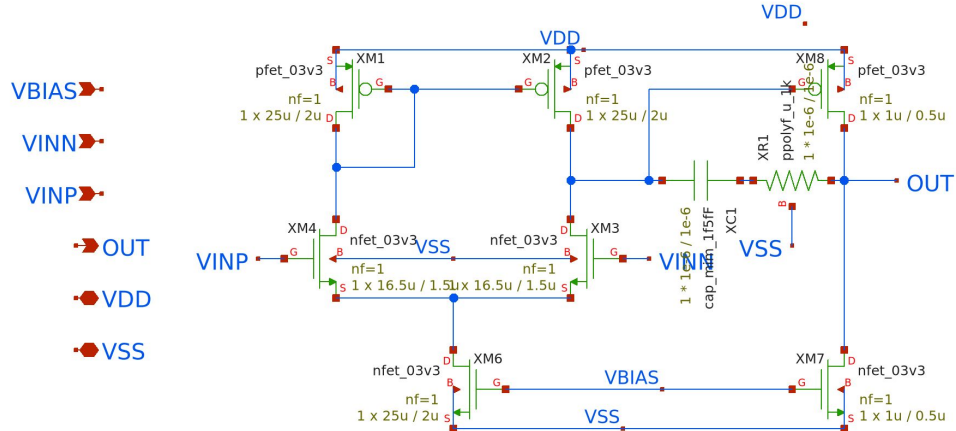
Inverter

M6_Low Noise Anomaly : CLB Internals – MUX2_1



Testbench schematic

M6_Low Noise Anomaly

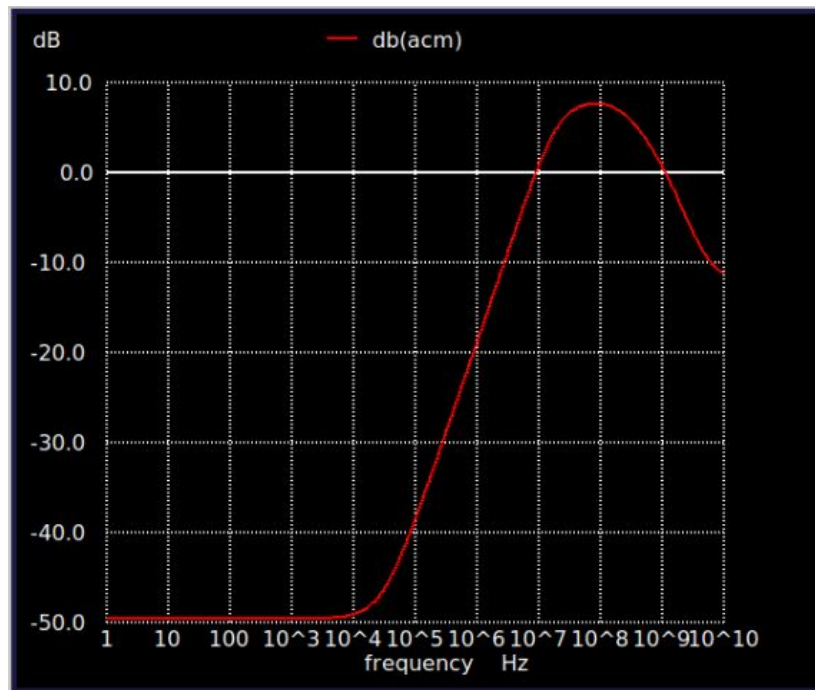


2-stage nmos OTA

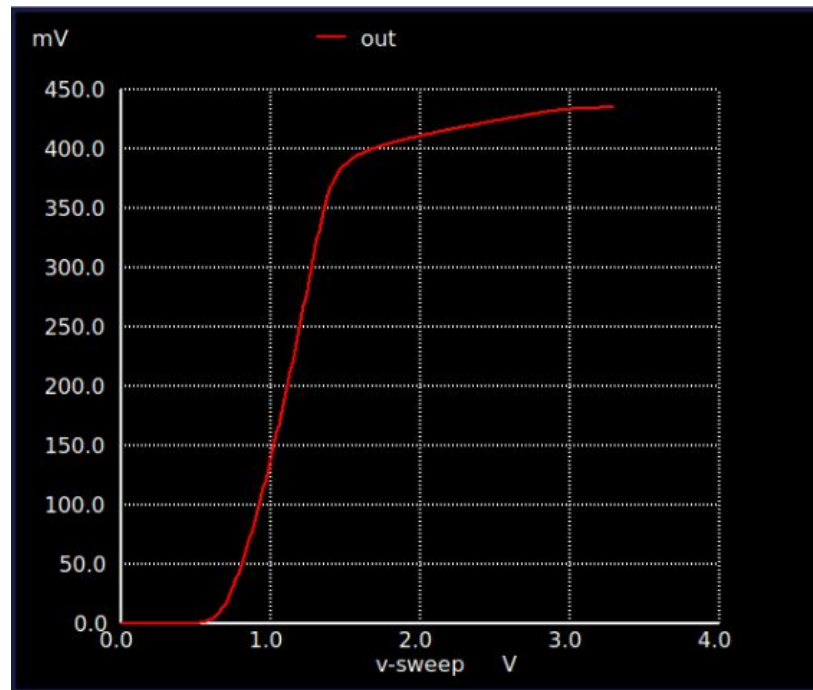
Operating Points

Corner Case	Typical
Ibias	10uA
VDD	3.3 Volts

M6_Low Noise Anomaly

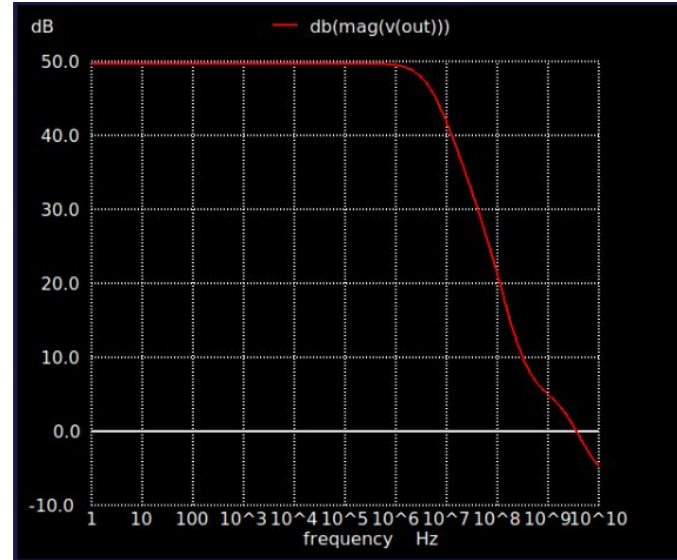
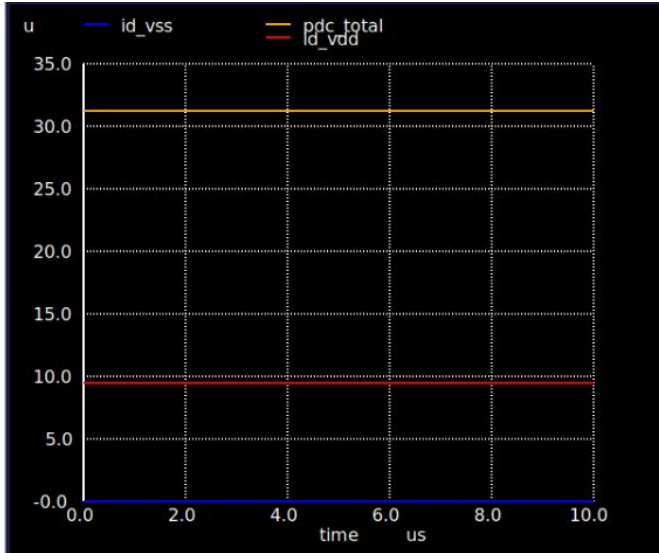


Common-Mode Gain



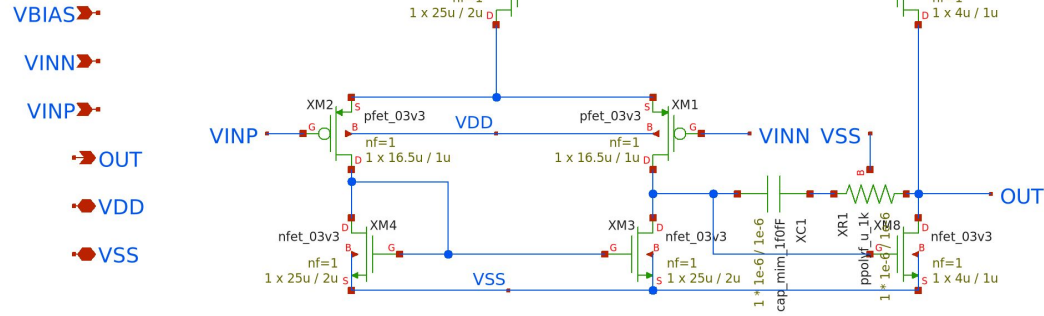
Input Common Mode Range

M6_Low Noise Anomaly

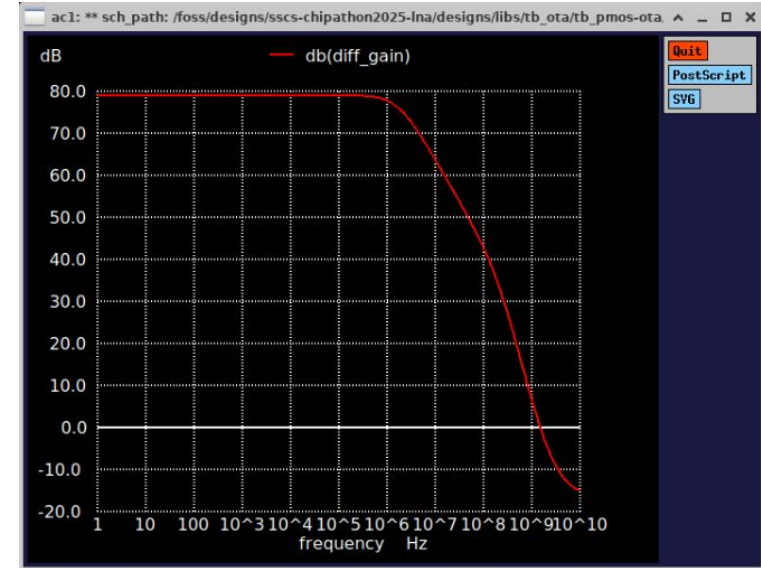


Power Consumption and PSRR

M6_Low Noise Anomaly



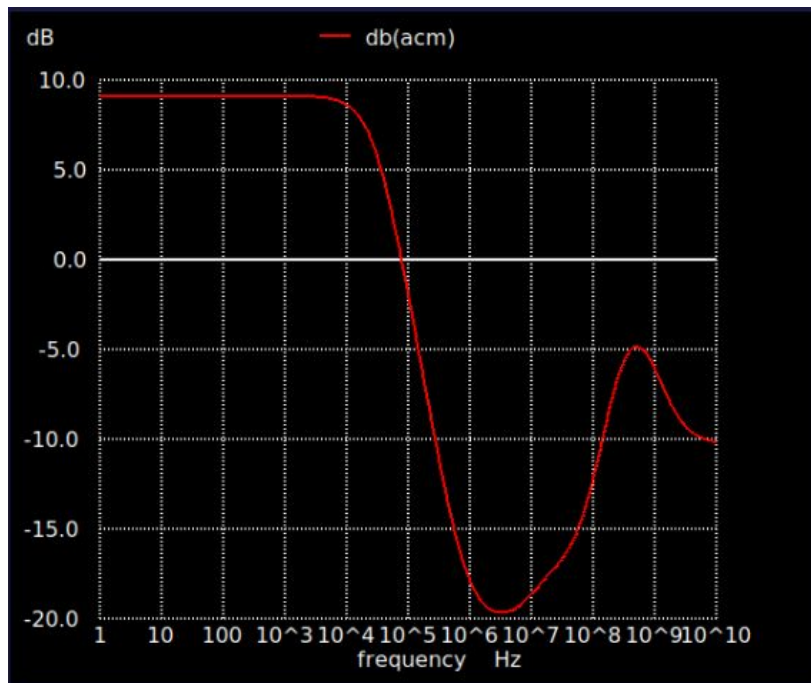
2-stage pmos OTA



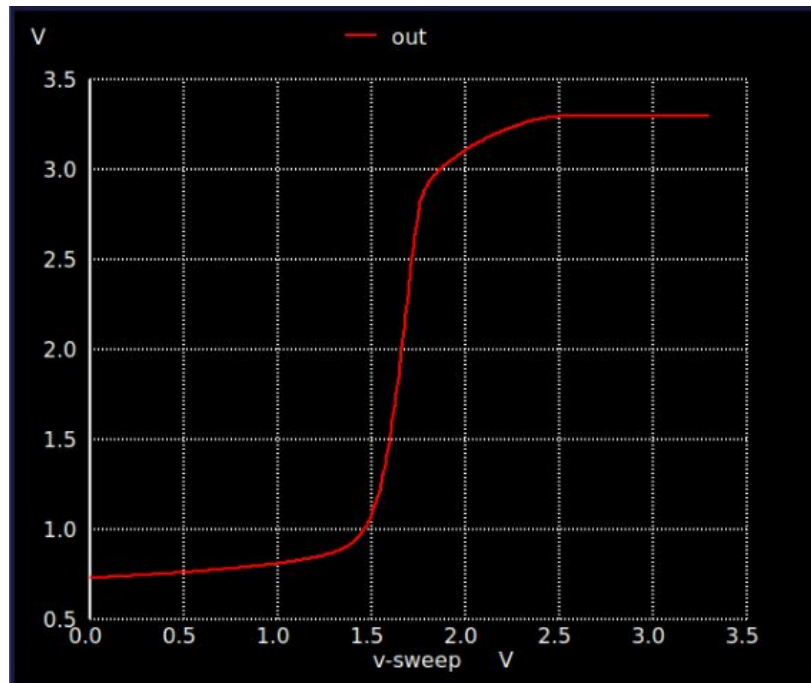
Operating Points

Corner Case	Typical
Ibias	10uA
VDD	3.3 Volts

M6_Low Noise Anomaly

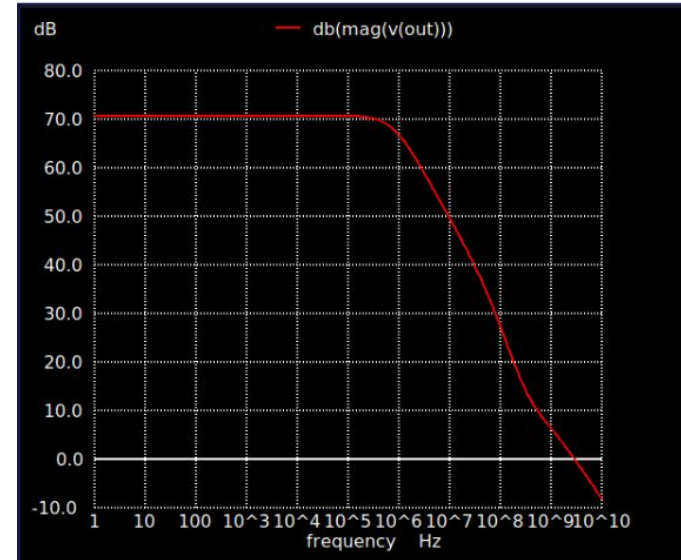
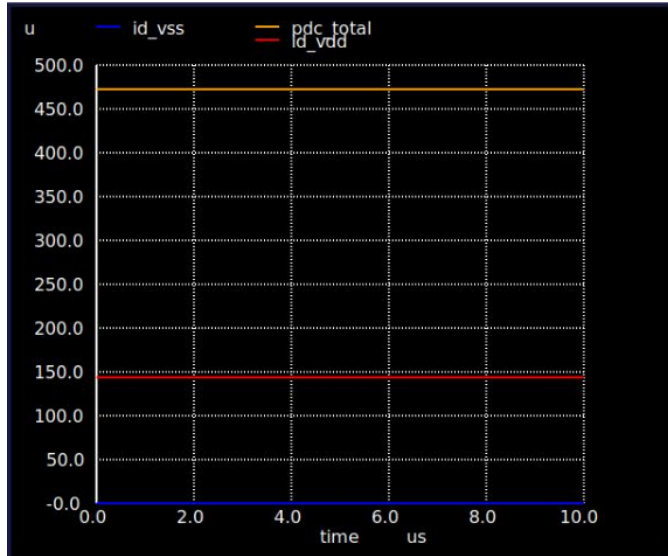


Common-Mode Gain



Input Common Mode Range

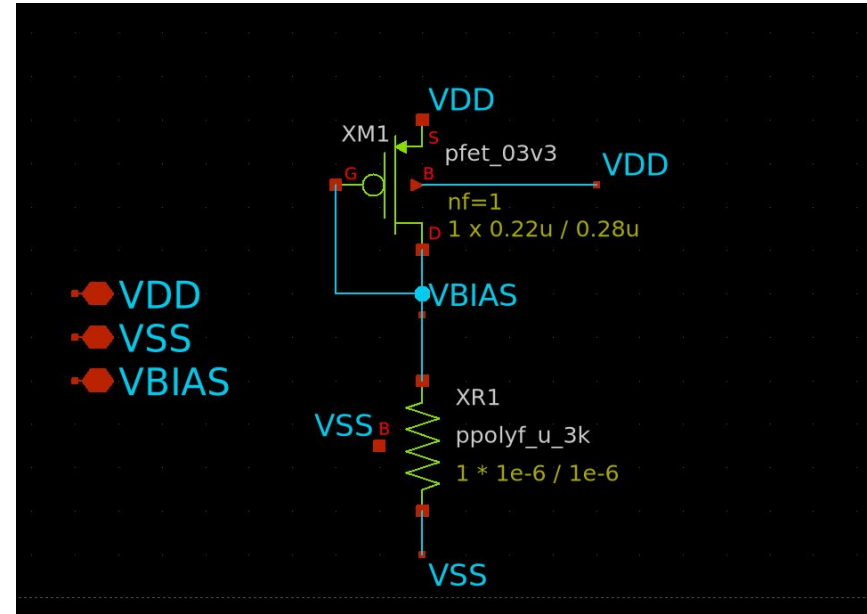
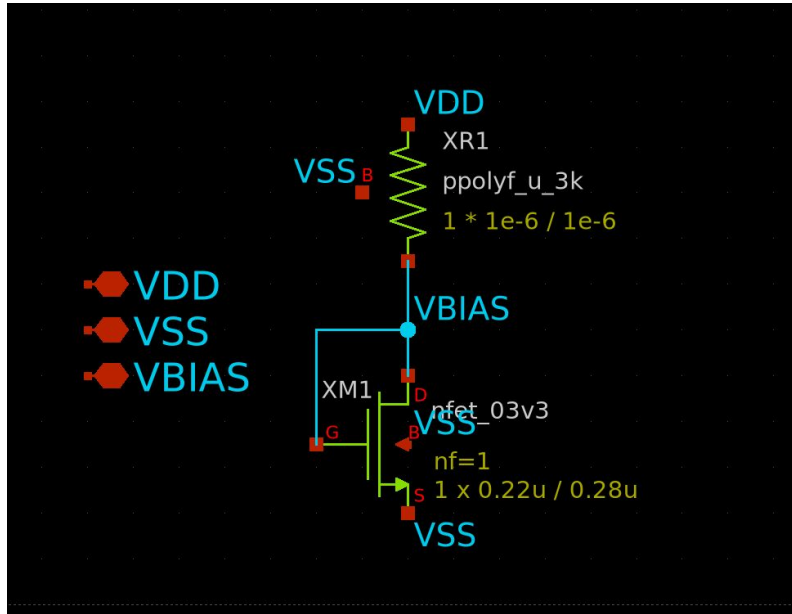
M6_Low Noise Anomaly



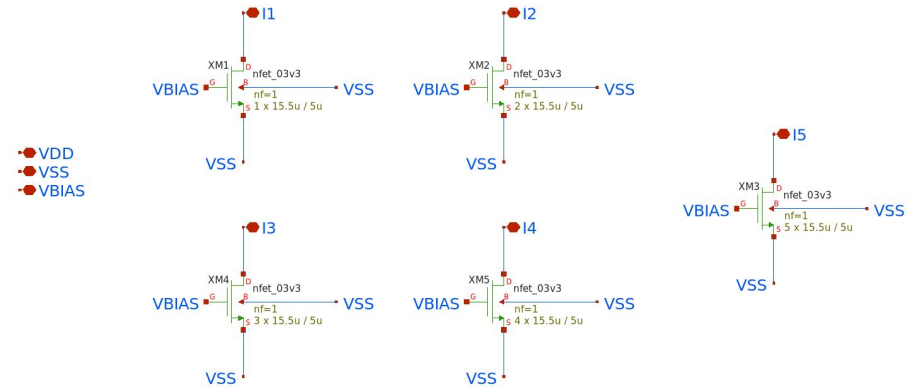
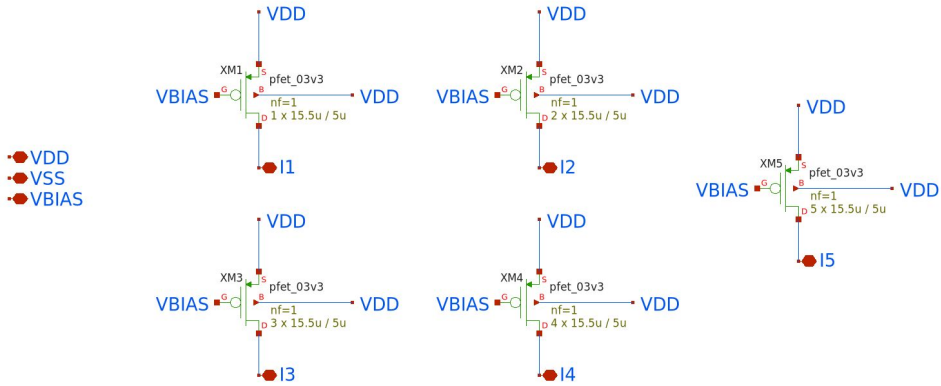
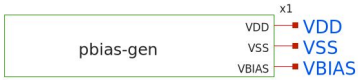
Power Consumption and PSRR

M6_Low Noise Anomaly

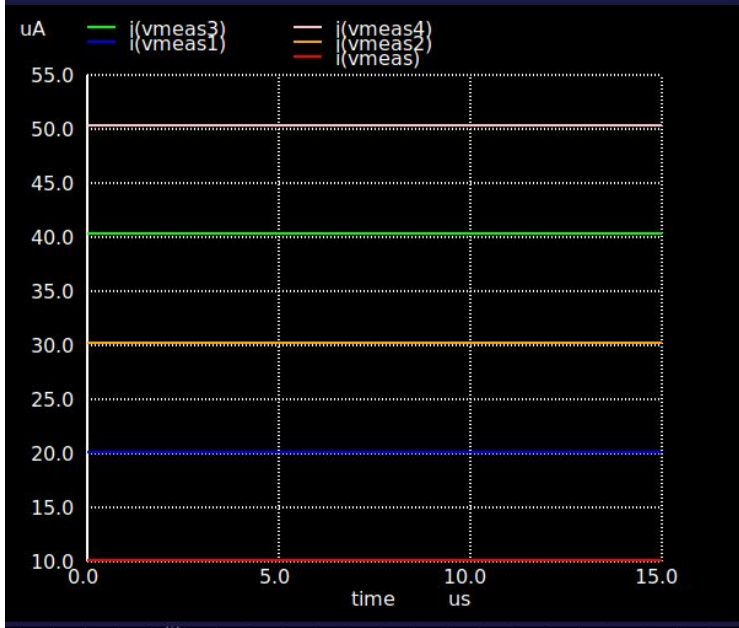
MOSFET Diode P and N



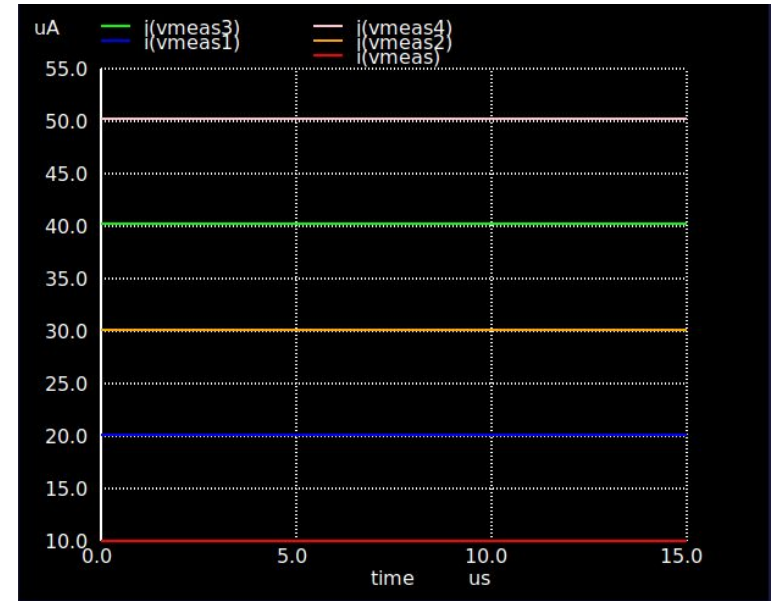
M6_Low Noise Anomaly - PMOS and NMOS Cascode Current Mirror



M6_Low Noise Anomaly - PMOS & NMOS Current Mirror Simulation



PFET Current Mirror



NFET Current Mirror